

 sentera SENSORS & DRONES RTK/PPK Units Production Guide	December 5, 2025
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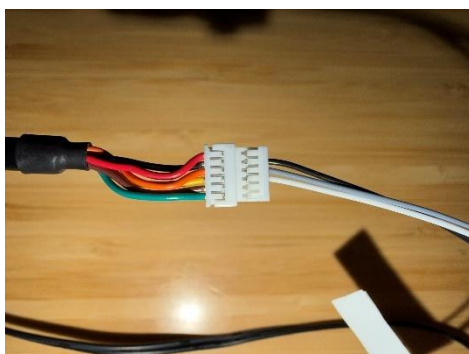
1. Equipment and Materials

In addition to the items listed in the bill of materials, the following are also needed:

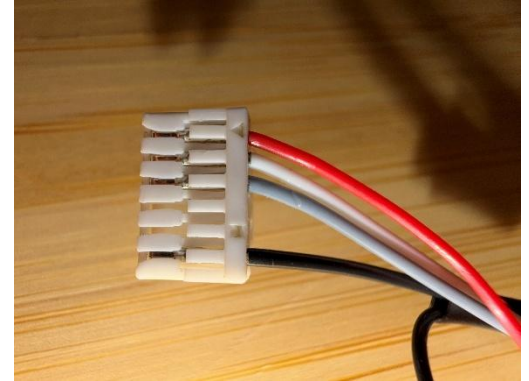
- Paper towels
- Scissors
- Araldite
- Mixing tray
- Stirring sticks
- Isopropyl alcohol
- Loctite
- Ruler
- Soldering Station (solder, iron, etc.)
- White 28 AWG wire
- JST crimp

2. Programing P900 Boards (for Air and Ground Modules)

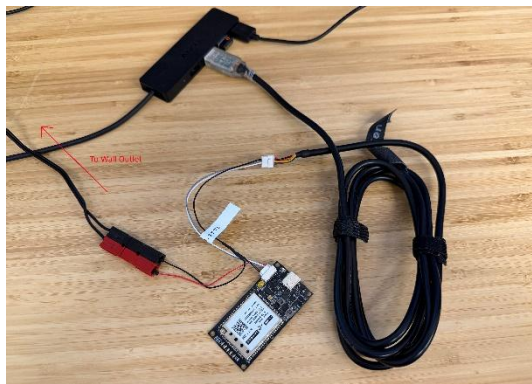
1. Open the 'Pico Config' Program.
 - a. On a computer, navigate to:
\\as-aurus.jdnet.deere.com\Data\Part Database (things we sell)\SW-32011 -- P900
Radio_tools\PicoConfig1.7
And open the 'PicoConfig' application.
2. Plug in the P900 board to the computer.
 - a. Use a TTL-232RG USB cable, found at Digi-Key:
<https://www.digikey.com/en/products/detail/ftdi-future-technology-devices-international-ltd/TTL-232RG-VSW3V3-WE/2441360>
 - b. Connect wires to JST connectors as shown.



- c. Connect power using an AC Power cord and direct into pins 1 (power) and 6 (ground).



- d. Plug in everything as shown.



3. Program the board.
- NOTE: if the light stutters/dims and turns off at any point in the programming process, refer to step 4.

- b. Set the values to the following NOTE: do not set the profile ('Load factory defaults') yet. The main goal of this step is to lower the power as often as possible.

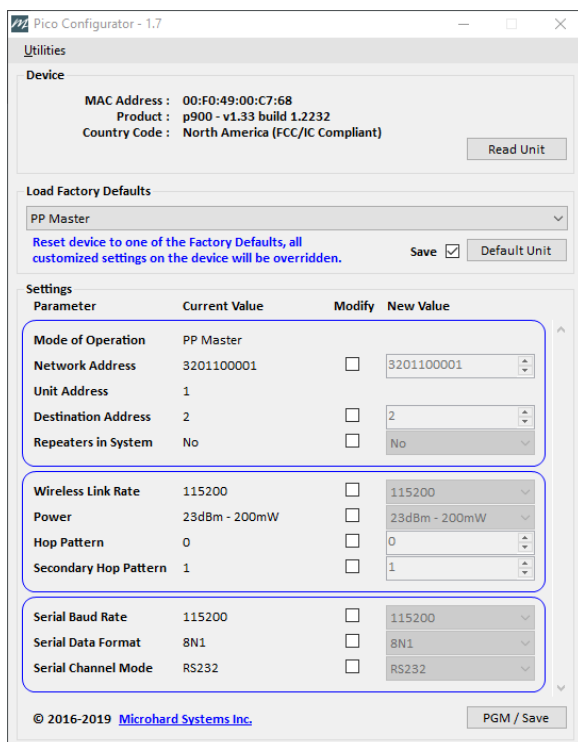


Figure 1: Use this configuration for Air (Master) Units.

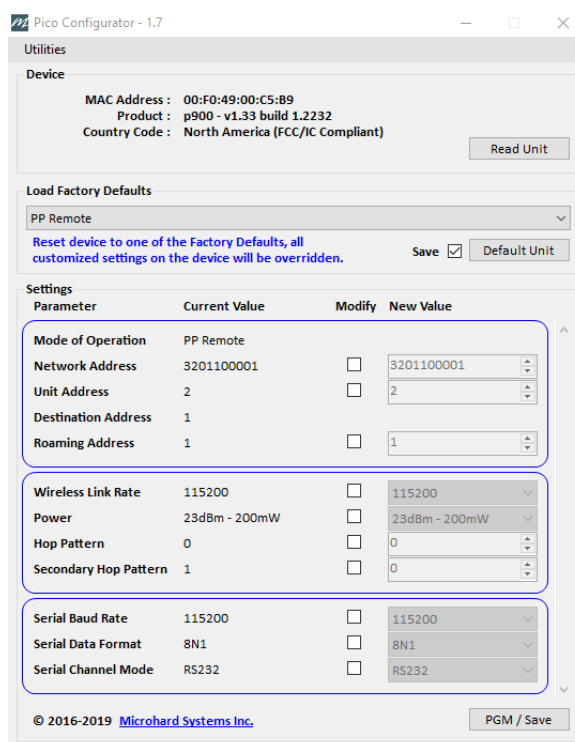


Figure 2: Use this configuration for Remote (Ground) Units.

- c. Select the correct option under 'Load Factory Defaults' and click 'Default Unit'.
- 'PP Master' for Air Units
 - 'PP Remote' for Ground Units
- d. Set the values again to the same as step 3b.
- NOTE: 'Network Address' should be set to the Remote Net ID intended for the air/ground units. It should be 32011XXXXX.
- e. Click 'Read Board' to double check the values have been set correctly, and the board itself has not been bricked (Light should still be on).

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4. FOR BRICKED BOARDS:

- a. If the board is bricked, the light will stutter/dim and turn off during programming. If this happens, unplug the board immediately and place it in a bag marked 'BAD'.

3. GPS Board Programing (for Air Modules)

1. Find the program.

- a. On a computer, navigate to
 \\as-aurus.jdnet.deere.com\Data\Part Database (things we sell)\SW-32014 -- ZED F9P\
- b. NOTE: You may refer to 'Instructions – IF800 RTK-PPK.md' for more information but it is not required.
- c. Copy 'ROVER6_A' file to local computer.
 - i. Desktop is fine, it is not permanent.
- d. NOTE: You may refer to 'README.md' in the 'ROVER6_A' file for more information but it is not required.

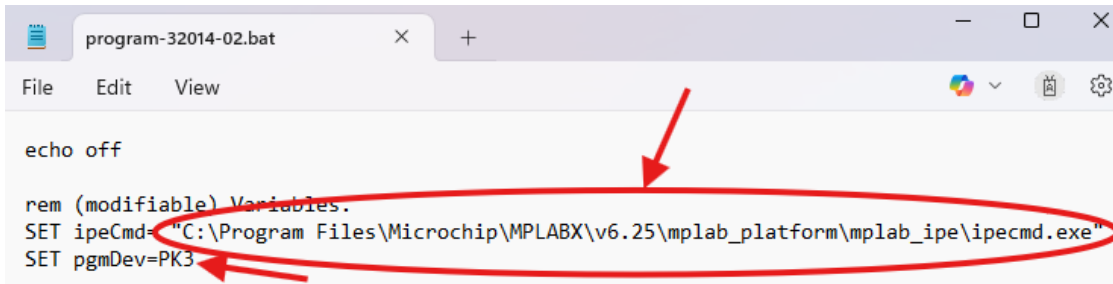
2. Install MPLAB X IDE/IPE

- a. Navigate to <https://www.microchip.com/en-us/tools-resources/develop/mplab-x-ide#tabs>
- b. Scroll down to downloads and click on the corresponding download.
 - i. Note: May need to download version 5.2. Other versions have proved to not work with our equipment.
- c. Once it has been downloaded, run the installer.
- d. Once you have installed MPLAB IPE. Run the application
- e. Update to newest version if prompted

3. Modify variables in the program.

- a. Open file explorer and navigate to the mplab-ipe file using the following path.
 - i. C:\Program Files\Microchip\MPLABX\v6.25\mplab_platform\mplab_ipe\
- b. Right Click on 'ipecmd.exe' and click 'Open as path'.
- c. Open the 'ROVER6_A' file on your computer.
- d. Right-click on 'program-32014-02' and click 'Edit in Notepad'.
- e. On the 4th line, you should find 'SET ipeCmd=...'. Delete the existing address and paste the path you just copied.
 - i. NOTE: make sure there is only 1 set of quotation marks. Another set will be pasted with the path.

- f. On the 5th line, you should find 'SET pgmDev='. Change this line to 'SET pgmDev=PK3'. This is to specify which programming device is being used. 'PICkit™ 3'



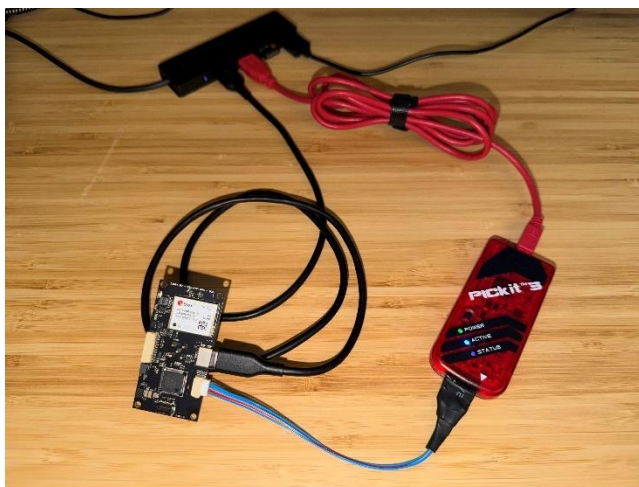
```
echo off

rem (modifiable) Variables.
SET ipeCmd= "C:\Program Files\Microchip\MPLABX\v6.25\mplab_platform\mplab_ipe\ipecmd.exe"
SET pgmDev=PK3
```

- g. Save and close notepad.

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4. Plug in the GPS Board
 - a. Plug the PICKit™ 3 board into a USB port on your computer.
 - b. Connect the JST comm connector to the GPS board.
 - c. Plug a USB-c cable into a USB port on your computer.
 - d. Connect the USB-c power cable to the GPS board.



5. Program the board.
 - a. Execute 'program-32014-02.bat'.
 - b. After the board is programmed, it will say 'Board program successful, press any key to exit'.
 - c. Check that TWO lights are flashing, one on either side of the board.
 - i. If one or both lights aren't flashing, disconnect everything and place the board back in its bag and mark 'BAD'.
6. Update the firmware.
 - a. For additional information on updating the firmware, go to:
<https://drotek.gitbook.io/rtk-f9p-positioning-solutions/tutorials/updating-zed-f9p-firmware>
In your browser. This is not required.
 - b. Navigate to <https://www.u-blox.com/en/product/zed-f9p-module?legacy=Current#Documentation-&-resources>
In your browser.
 - c. Scroll down and click 'Documentation & Resources'. Click 'Firmware Update'. And download the latest version. Place this file in your 'ROVER6_A' folder.
 - d. Navigate to and start the U-block 'u-center' software.
 - e. In the top left corner, click on the upside-down triangle and select the COM port related to the GPS board.

- f. Wait for the connection icon to turn green. This indicates the serial connection to the GPS board is working.
- g. At the top, click 'Tools' > 'Firmware Update...' to open the firmware update utility window.
 - i. Click on the 3 dots next to the Firmware Image.
 - ii. Find and select the firmware update file that you moved to the 'ROVER6_A' folder.
 - iii. Click 'open'
 - iv. Make sure the option 'Use this baud rate for update' is checked and the baud rate is set to 9600 bauds.
 - v. Initiate the firmware update by clicking on the green 'GO' at the bottom of the window.
 - vi. Once the update process is done, a message saying 'Firmware Update Utility completed successfully' will be displayed in the console box. NOTE: you may have to enlarge the window to view all the text in the console box.
- h. If you want to verify the update version number, at the top, select 'View' > 'Messages View'.
 - i. On the tree structure, click 'UBX' > 'MON (Monitor)' > 'VER (Version)'. The right-side panel will list the ZED-F9P version number.

4. Sticker Printing

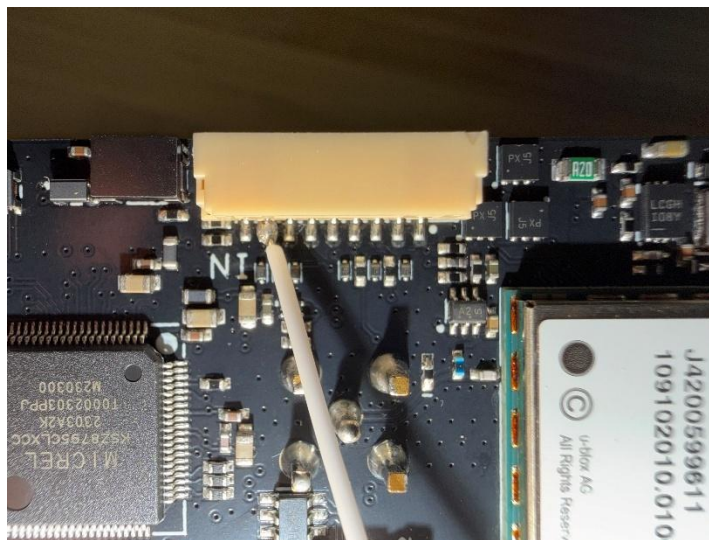
1. Note the radio net ID numbers for both the air and ground radio boards. They should be the same.
2. Note the serial numbers that should be used for the air modules.
3. Print the labels using current templates on the label computer.
 - a. There should be 2 of each radio net ID and 1 of each serial number.



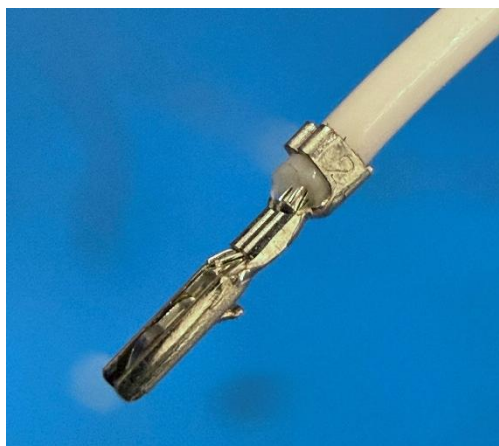
5. Air Module (21077-XX) Assembly

Note: It may be easiest to use the 3D-printed enclosure cradle while installing components into enclosure to avoid tipping.

1. Solder a 4' white wire to the pin on the back of the connector of the GPS board.
 - a. Measure and cut 6" of white 28 AWG wire.
 - b. Strip one end and solder to pin #8 of the 9-pin JST connector marked 'IN' on the GPS board.

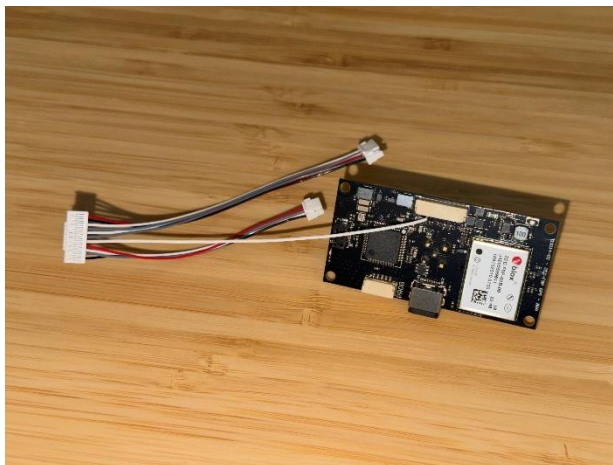


- c. Measure and cut wire to 4".
 - d. Strip wire end and crimp a JST connector onto it.



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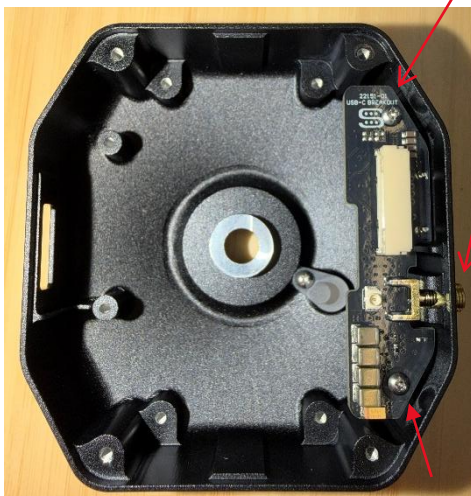
- e. Insert JST crimp into pin 7 of the 11-pin connector in wire.



2. Install diffuser to enclosure using screw (item #11 in drawing) and Loctite.



3. Install USB-C Breakout Board.
 - a. Remove nut and lock washer from USB-C breakout board MCX connector. Discard lock washer.
 - b. Place board into position in enclosure. Loosely fasten nut on MCX connector. Install 2 screws (item #11 in drawing) with Loctite.
 - c. Tighten nut with wrench.



4. Install GPS board.

- a. Plug the 8-pin JST connector into the ETH2 port of the GPS board.



- b. Remove nut and lock washer from SMA connector on GPS board. DO NOT DISCARD.
c. Place GPS board into enclosure. GPS board should be oriented such that the LED aligns with the diffuser and all 4 screw holes are aligned with those on the enclosure.

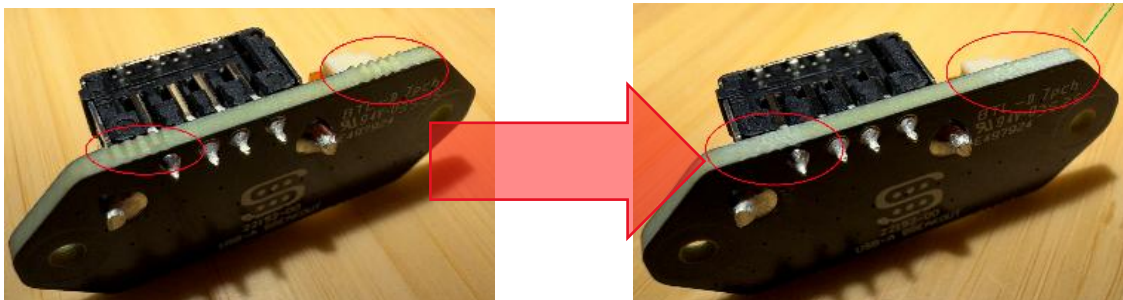


- d. Replace lock washer and nut on SMA connector. Tighten nut with wrench. Double check the screw holes are still aligned.



5. Install USB-A breakout board.

- a. Carefully file mouse bites off USB-A breakout board.

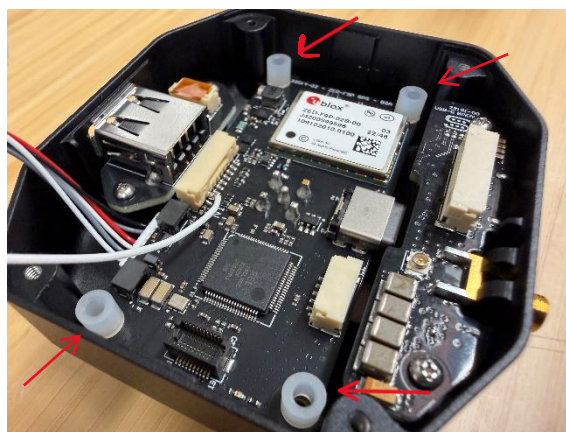


- b. Install USB-A breakout board to enclosure using 2 screws (item #11 in drawing) and Loctite.



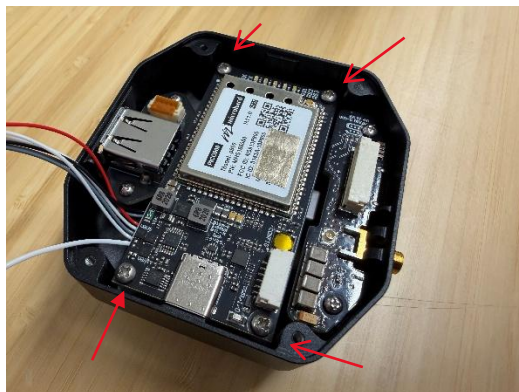
6. Install radio board.

- a. Carefully place 4 spacers on GPS board above screw holes.



- b. Install radio board on top of GPS board making sure to align the connector on the back.

- c. Install 4 screws (item #10 in drawing) through radio board, spacers, and GPS board with Loctite.



7. Connect the cable to the USB-C and USB-A breakout boards and route wires as shown.



8. Install the coax cable to the radio and USB-C boards and route as shown.



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9. Prepare and install cover.

- a. Cut 2 small squares from the foam pad and install them on the raised circles of the cover.



- b. Install the cover to the enclosure using 4 screws (item #12 in drawing) and Loctite.



10. Install Antenna.

- a. Install the O-ring into the channel on top of the enclosure.



- b. Install the helical antenna to the SMA connector on top of the enclosure. Tighten by hand until snug.



11. Install 4 captive thumb screws into threaded holes of the cover.



12. Clean empty side of enclosure with alcohol. Apply Radio Net ID sticker with corresponding number as shown.

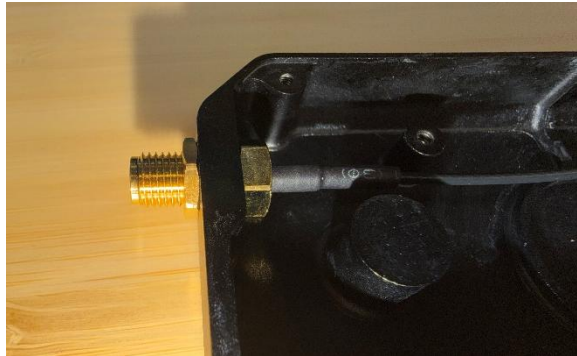


13. Clean 'Sentera' side of enclosure with alcohol. Apply Serial number sticker with corresponding numbers under logo as shown.



6. Ground Module (31039-XX) Assembly

1. Install RF cable.
 - a. Remove locking washer and nut from cable. Discard locking washer.
 - b. Install RF cable and tighten nut with wrench.



- c. Apply Araldite 2-part adhesive to rear side of SMA connector as shown.
 - i. Note: This adhesive is intended to prevent the connector from spinning when installing/uninstalling antenna.
 - ii. Note: Cure time is 24 hours, but it is not required to wait until cured to continue assembly.



2. Assemble and install magnet stack
 - a. Clean both sides of magnet with alcohol.
 - b. Apply mounting tape circle to one side and foam bumper to other side.
 - i. Note: Magnet is not directional, so the sides do not matter.



- c. Clean round recess inside enclosure with alcohol.
- d. Install magnet stack into enclosure with foam bumper facing upward.



- 3. Loosely install USB-C breakout board.
 - a. Use 2 Shim washers (item #9 in drawing), 2 Washers (item #10 in drawing), and 2 screws (item #11 in drawing) to install the USB-C breakout board into enclosure with Loctite.
 - i. Note: it is easiest to insert board into place. Then place shim washers on the board around the enclosure standoffs, then install screws with washers.
 - ii. Note: Tighten screws but barely not enough to prevent board from shifting. If the pressure is correct, the board may move but friction from screws will prevent it from moving unintentionally.



4. Align and install radio board.
 - a. Install the P900 board to the USB-C breakout board and shift the boards such that ALL 4 of the mounting holes align between enclosure and P900 board.



- b. Press on USB-C connector of the USB-C breakout board to prevent shifting and remove the P900 board.
 - i. Note: it may be easiest to pull on the USB-C connector side of the P900 board because the connector on the bottom is on that side. This will prevent twisting/shifting.



- c. Tighten screws to install USB-C breakout board completely.
 - d. Route the coax cable as shown.



- e. Re-install the P900 board to USB-C board and make sure the holes are still aligned. If not, remove P900 board, loosen USB-C breakout board screws and repeat step 4.
 - f. Make sure nothing is squishing the coax cable by pressing on the P900 board and sliding the coax cable back and forth. There should be no resistance.

- g. Install 4 screws (item #12 in drawing) using Loctite.



5. Plug the coax cable into the P900 board and route as shown.



6. Cut a small square out of a foam bumper and stick it to the raised circle on the cover. This foam bumper ensures the RF cable stays connected during use.



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7. Apply stickers.
 - a. Clean the recessed area of the cover with alcohol.
 - b. Apply the label to the cover ensuring the transparent circles on the label align with the holes of the cover.
 - i. Note: it may be easiest to use 2 tweezers on either side of the sticker to enable finer control of the placement.
 - c. Clean the cover with alcohol.
 - d. Apply the radio net ID sticker with the corresponding number as shown.



8. Install the cover to the enclosure using 4 screws (item #13 in drawing) and Loctite.



9. Install felt sheet.
 - a. Use the plastic template to cut the adhesive-backed felt into proper shape.
 - b. Clean bottom side of enclosure (the side with the threaded hole) with alcohol.
 - c. Apply felt sheet to enclosure making sure to align threaded hole in the middle of the hole in the felt sheet.



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10. Install the antenna to the radio module and hand tighten.



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7. Case (21076-XX) Assembly

1. Apply label to outside of case.
2. Assemble equipment into case as shown. Loose items in large pocket should be placed together into a clear Ziplock bag.
 - a. IF800



b. IF1200



c. asdf